

Phil Steer's Research Using the SMMIS Dataset: A Comprehensive Literature Review

Overview

Professor Philip J. Steer (Emeritus Professor of Obstetrics and Gynaecology, Imperial College London, Chelsea and Westminster Hospital) has produced an extensive body of research using the St Mary's Maternity Information System (SMMIS) — a computerised maternity records database covering pregnancies booked at 15–18 hospitals in the former North West Thames Regional Health Authority area in London, from 1988 to 2000 inclusive. The system recorded detailed clinical, demographic, obstetric, and neonatal data for every pregnancy, ultimately covering 585,291 pregnancies. Data were entered prospectively by trained midwives and clerks with online validation, prompting, and standardised clinical definitions, producing a high-quality longitudinal dataset.^{[1][2][3][4][5]}

The SMMIS database has been described as uniquely valuable for studying pregnancy outcomes due to its large size, regional coherence, ethnic diversity of the population, and the range of both antenatal and intrapartum variables it contains. Phil Steer served as one of the principal custodians of the dataset and as either lead or named co-author on numerous studies drawing from it. The research spans maternal haemoglobin, blood pressure, gestational diabetes, preterm birth seasonality, meconium-stained amniotic fluid, intrapartum risk factors, and neonatal adverse outcomes.^{[6][3]}

Study 1: Maternal Haemoglobin and Birth Weight Across Ethnic Groups (1995)

Reference: Steer P, Alam MA, Wadsworth J, Welch A. *Relation between maternal haemoglobin concentration and birth weight in different ethnic groups*. BMJ 1995; 310(6978):489–491.^{[7][8]}

Dataset size: 153,602 pregnancies (1988–1991), with ethnic group and birth weight recorded.

Key Findings

- Maximum mean birth weight in white women was achieved when the lowest haemoglobin (Hb) concentration during pregnancy was between 85–95 g/L; the lowest incidence of low birth weight and preterm labour occurred with a lowest Hb of 95–105 g/L.^{[9][7]}
- A similar U-shaped pattern of birth weight against lowest Hb was observed across all ethnic groups (white European, Indo-Pakistani, Afro-Caribbean, Mediterranean, black African, Oriental).^[^7]
- Failure of Hb to fall below 105 g/L is a marker of inadequate plasma volume expansion and is independently associated with an increased risk of low birth weight and preterm delivery — across all ethnic groups.^{[9][7]}

- Ethnic minorities (particularly Indo-Pakistani and black African women) had higher rates of both low Hb and low birth weight, but the raised rate of low birth weight in these groups was not explained by their higher rates of "anaemia" — the U-shaped relationship between Hb and birth weight existed independently.^[8]^[7]
- The study established that Hb in pregnancy should be treated as a continuous variable (not dichotomised as "anaemic" or "non-anaemic"), because the relationship with birth weight is non-linear.^[^10]

Study 2: Birth Weight — Nature or Nurture? (1995)

Reference: Brooks AA, Johnson MR, Steer PJ, Pawson ME, Abdalla HI. *Birth weight: nature or nurture?* Early Human Development 1995; 42(1):29–35.^[11]^[12]

Dataset: 62 cases of ovum donation (distinct from SMMIS but contextually linked to Steer's group at Chelsea and Westminster).

Key Findings

- The environment provided by the mother (recipient weight and gestational age) was the dominant determinant of birth weight following ovum donation — not donor genetics.^[^12]
- Donor weight, the donor's own birth weight, and the birth weight of the donor's own children showed no significant correlation with the birth weight of the donated-egg infant.^[^12]
- This provided early mechanistic context for the SMMIS population findings on birth weight determinants — that maternal physiological factors, rather than fetal genetic endowment alone, are paramount.

Study 3: Maternal Blood Pressure, Birth Weight, and Perinatal Mortality (2004)

Reference: Steer PJ, Little MP, Kold-Jensen T, Chapple J, Elliott P. *Maternal blood pressure in pregnancy, birth weight, and perinatal mortality in first births: prospective study.* BMJ 2004; 329(7478):1312.^[3]^[11]

Dataset size: 210,814 first singleton births from 15 maternity units, 1988–2000.

Key Findings

- Birth weight was maximal when the highest recorded diastolic blood pressure during pregnancy was between 70–90 mmHg, following an inverted U-shaped relationship.^[^3]
- Both low and high diastolic blood pressure were associated with small babies and elevated perinatal mortality — not only hypertension, as previously assumed.^[^3]
- Using a linear-quadratic model, 94 of 824 perinatal deaths (11.4%) were attributable to mothers with blood pressure differing from the model-optimal level of 82.7 mmHg; the

majority (91.2%) of these excess deaths occurred in women with blood pressures *below* the optimal value.^[^3]

- Low blood pressure at booking (≤ 70 mmHg) combined with a rise of 15–30 mmHg during pregnancy was *beneficial* for birth weight, whereas a rise exceeding 30 mmHg was harmful — demonstrating that the **direction and magnitude of change** in blood pressure matters, not just the absolute level.^[^3]
- The third-trimester physiological rise in blood pressure appears to be an adaptive response to support placental perfusion and fetal growth.^[^3]
- Perinatal mortality remained curvilinearly associated with blood pressure even after adjustment for birth weight, ethnic group, smoking, BMI, deprivation, and other confounders.^[^3]

Study 4: Maternal Haemoglobin, Perinatal Mortality (2005)

Reference: Little MP, Brocard P, Elliott P, Steer PJ. *Hemoglobin concentration in pregnancy and perinatal mortality: a London-based cohort study*. American Journal of Obstetrics & Gynecology 2005;193(1):220–226.^{[13][14]}

Dataset size: 222,614 first singleton pregnancies from the SMMIS database.

Key Findings

- Haemoglobin at the first antenatal check showed no statistically significant association with perinatal mortality.^[^13]
- A statistically significant U-shaped relationship ($P < 0.001$) was found between the *lowest recorded* maternal Hb concentration during pregnancy and perinatal mortality: both very low and relatively high Hb concentrations were associated with increased risk.^[^13]
- Lowest perinatal mortality was associated with a lowest recorded maternal Hb between 9–11 g/dL (90–110 g/L).^{[1][13]}
- The association between lowest Hb and early neonatal mortality was largely mediated by preterm birth; the association between lowest Hb and stillbirth was largely mediated by fetal growth restriction.^[^13]
- This confirmed that the optimal Hb range in pregnancy is well below the conventional "non-anaemic" threshold, and that the plasma volume expansion reflecting healthy fetal growth is itself protective.^{[15][13]}

Study 5: Maternal Haemoglobin and Birth Weight — Summary Review (2000)

Reference: Steer PJ. *Maternal hemoglobin concentration and birth weight*. American Journal of Clinical Nutrition 2000;71(5 Suppl):1285S–1287S.^{[11][15]}

Key Findings

- Synthesised the SMMIS findings alongside other evidence: **severe anaemia** (Hb <80 g/L) is associated with small babies through both preterm labour and growth restriction.[¹⁵]
- **Failure of plasma volume to expand** (reflected by persistently high Hb >120 g/L at end of second trimester) is associated with up to a 3-fold increased risk of pre-eclampsia and intrauterine growth restriction.[¹⁵]
- A mean corpuscular volume (MCV) <84 fL in association with Hb <95 g/L is a reliable indicator of iron deficiency rather than physiological dilution.[¹⁵]
- The minimum incidence of low birth weight (<2.5 kg) and preterm labour (<37 weeks) occurs with a lowest Hb in the range of 95–105 g/L — the narrow physiological optimum identified from the SMMIS population.[¹⁵]

Study 6: Gestation and Ethnicity (2004)

Reference: Patel RR, Steer P, Doyle P, Little MP, Elliott P. *Does gestation vary by ethnic group? A London-based study of over 122,000 pregnancies with spontaneous onset of labour*. International Journal of Epidemiology 2004; 33(1):107–113.[¹⁶][¹⁷][¹⁸]

Dataset size: 122,415 nulliparous women with singleton live fetuses in spontaneous labour.

Key Findings

- **Median gestational age at delivery** was 39 weeks for Black and Asian women, compared with 40 weeks for white European women.[¹⁶]
- Black women with normal BMI had significantly increased odds of preterm delivery (OR 1.33, 95% CI 1.15–1.56, adjusted for deprivation and BMI) compared with white Europeans. [¹⁶]
- Asian women also had increased odds of preterm delivery (OR 1.45, 95% CI 1.33–1.56).[¹⁶]
- **Meconium-stained amniotic fluid** — an indicator of fetal maturity — was statistically significantly more frequent in preterm Black and Asian infants, and in term Black infants, compared with white European infants.[¹⁶]
- The study concluded that **normal gestational length is biologically shorter** in Black and Asian women compared with white European women, and that fetal maturation may occur earlier in these groups.[¹⁸][¹⁶]
- These findings have important implications for gestational age-based clinical guidelines and population-specific norms.

Study 7: Seasonal Patterns and Preterm Birth (2006)

Reference: Lee SJ, Steer PJ, Filippi V. *Seasonal patterns and preterm birth: a systematic review of the literature and an analysis in a London-based cohort*. BJOG 2006; 113(11):1280–1288.[¹⁹][²⁰][²¹]

Dataset size: 482,765 live singleton births after 24 weeks' gestation from SMMIS (1988–2000).

Key Findings

- Preterm births in the SMMIS London cohort peaked only once a year — in winter — unlike other developed countries (USA, Australia) where a bimodal (winter and summer) pattern is more common.[²⁰][¹⁹]
- Babies born in winter were 10% more likely to be preterm than those born in spring (OR 1.10, 95% CI 1.07–1.14).[¹⁹][²⁰]
- This unique single-peak London pattern was not explained by air pollution or meteorological factors (see Study 8 below), suggesting other biological or social mechanisms specific to the London/UK population.[²⁰]
- Establishing a seasonal pattern of preterm birth has direct implications for maternity service capacity planning and neonatal care resource allocation.[²⁰]

Study 8: Air Pollution and Meteorological Factors in Preterm Birth (2008)

Reference: Lee SJ, Hajat S, Steer PJ, Filippi V. *A time-series analysis of any short-term effects of meteorological and air pollution factors on preterm births in London, UK*. Environmental Research 2008; 106(2):185–194.[²²][²³]

Dataset size: 482,568 births from SMMIS (1988–2000), linked with daily environmental data.

Key Findings

- No significant association was found between ambient air pollution (ozone, PM10) and risk of preterm birth in London over the 13-year study period.[²²]
- No significant association was found between any meteorological factor (temperature, rainfall, sunshine, relative humidity, barometric pressure, or largest daily drop in barometric pressure) and preterm birth.[²²]
- Cumulative exposure up to 6 days before birth was also not significantly associated with preterm birth risk.[²²]
- This large-scale null finding is significant: it rules out short-term weather or pollution triggers for the London winter preterm birth peak identified in Study 7, focusing attention on longer-term or biological explanations.[²²]

Study 9: Gestational Diabetes and Ethnicity (2012)

Reference: Makgoba M, Savvidou MD, Steer PJ. *An analysis of the interrelationship between maternal age, body mass index and racial origin in the development of gestational diabetes mellitus*. BJOG 2012; 119(3):276–282.^[24]^[11]

Dataset size: 1,688 women who developed GDM and 172,632 who did not (all nulliparous); 15 maternity units, 1988–2000.

Key Findings

- Both advancing maternal age and increasing BMI were independently and strongly associated with development of GDM ($P < 0.01$ for both), and these associations were modified by racial group.^[^24]
- Compared with white European women aged 20–24 years, significantly elevated GDM odds were found in white Europeans only from age >30 , but in Black African women from age >25 and in South Asian women from age >20 .^[^24]
- Black African and South Asian women had significantly higher odds of GDM irrespective of BMI — meaning their elevated risk cannot be attributed to obesity alone.^[^24]
- The study demonstrated that GDM screening criteria and thresholds may need to be ethnicity-specific, particularly with regard to the age at which to begin offering glucose tolerance testing.^[^24]

Study 10: Meconium-Stained Amniotic Fluid — Characteristics and Risk Factors (2011)

Reference: Balchin I, Whittaker JC, Lamont RF, Steer PJ. *Maternal and fetal characteristics associated with meconium-stained amniotic fluid*. Obstetrics & Gynecology 2011; 117(4):828–835.^[25]^[26]

Dataset size: 499,096 singleton births ≥ 500 g at 24+ weeks, SMMIS (1988–2000).

Key Findings

- Meconium-stained amniotic fluid (MSAF) rates increased markedly with gestation: 5.1% preterm, 16.5% at term, and 27.1% post-term.^[^25]
- MSAF rates varied substantially by ethnicity: 22.6% in Black women, 16.8% in South Asian women, and 15.7% in white women.^[^25]
- Independent predictors of MSAF included: being Black (OR 8.4, 95% CI 2.4–28.8), vaginal breech delivery (OR 4.7), being South Asian (OR 3.3), and advancing gestational age (OR 1.39 per week).^[^25]
- More Black (17.9%) and South Asian (11.8%) women with good outcomes and no risk factors for fetal hypoxia had MSAF than did white women (11.2%) — indicating that MSAF is more likely to be physiological (maturity-related) rather than pathological in Black and South Asian populations.^[^25]

- Using white neonates born at 40 weeks as reference, the absolute risk of adverse outcome at 41–42 weeks was 2–5% in white, 3–7% in South Asian, and 7–11% in Black infants — the highest absolute risk.[^25]
- MSAF rates are driven by both gestational age and ethnicity, and should be interpreted differently across racial groups when assessing fetal risk.[^25]

Study 11: Risk Factors for Serious Adverse Neonatal Outcome at Term (2023)

Reference: Jindal S, Steer PJ, Savvidou M, Draycott T, Dixon-Woods M, Wood A, Kim LG. *Risk factors for a serious adverse outcome in neonates: a retrospective cohort study of vaginal births*. BJOG 2023; 130(12):1521–1530.[^4][5]

Dataset size: 302,137 vaginal births at 37–42 weeks from SMMIS (585,291 pregnancies in total, 1988–2000).

Key Findings

Antenatal risk factors independently associated with 5-minute Apgar score <7:

- Suspected fetal growth restriction (FGR): OR 1.34 (95% CI 1.16–1.53)[^5]
- Induction of labour: OR 1.41 (95% CI 1.25–1.58)[^5]
- Nulliparity: OR 1.48 (95% CI 1.34–1.63)[^5]
- Booking BMI ≥ 30 : OR 1.18 (95% CI 1.02–1.37)[^5]
- Maternal age <25: OR 1.23 (95% CI 1.10–1.39)[^5]
- Black ethnicity: OR 1.21 (95% CI 1.03–1.43)[^5]
- Early-term birth (37–38 weeks): OR 1.13; late-term birth (41–42 weeks): OR 1.14[^5]

Intrapartum risk factors:

- Oxytocin use: OR 1.27[^5]
- Maternal pyrexia: OR 1.87 (95% CI 1.46–2.40) — almost double the risk[^5]
- Abnormal fetal heart rate pattern (aFHRp) without meconium: OR 2.40[^5]
- Meconium without aFHRp: OR 2.20[^5]
- Both aFHRp and meconium combined: OR 4.26 (95% CI 3.74–4.87) — a fourfold risk[^5]

For the composite adverse outcome (Apgar <7, intubation, or perinatal death):

- Meconium alone was associated with a 10-fold higher risk (OR 10.0)[^5]
- Both meconium and aFHRp together: 15-fold higher risk (OR 15.0, 95% CI 13.9–16.3)[^5]

Overarching conclusions:

- Monitoring fetal heart rate pattern alone is insufficient for assessing intrapartum fetal wellbeing; a full multi-factor clinical picture is required.^[^5]
- The study argued for development of a composite real-time risk score incorporating multiple factors, potentially using artificial intelligence, to improve intrapartum decision-making.^[^5]
- Black ethnicity was independently associated with adverse outcome even after adjusting for FGR and meconium — highlighting significant ethnic health inequalities in maternity care.^[^5]

Study 12: Intrapartum Interaction of Risk Factors and Adverse Labour Outcome (2023)

Reference: Steer PJ, Yau CWH, Blott M, Lattey K, Nwandison M, Uddin Z, Winter C, Draycott T. *A case-control study of the interaction of fetal heart rate abnormalities, fetal growth restriction, meconium in the amniotic fluid and tachysystole, in relation to the outcome of labour*. BJOG 2023; 130(3):286–291.^{[27][28]}

Design: Case-control study at 26 maternity units in the UK; 69 adverse outcome labours vs 198 consecutive control labours without adverse outcome.

Key Findings

- A risk score combining suspected FGR, tachysystole (excessive uterine contractions), meconium in amniotic fluid, and fetal heart rate abnormalities (baseline rate and variability, decelerations) was evaluated across hourly assessments for 7 hours before birth.^[^27]
- The combined risk score was markedly higher in adverse outcome labours than controls, confirming the critical importance of tracking multiple risk factors across time, not simply measuring them at a single point.^[^27]
- Tachysystole was identified as a clinically important but often overlooked risk factor in combination with FHR abnormalities and meconium.^[^27]
- This study directly extended and validated the SMMIS-derived findings of Study 11, applying them in a prospective multi-centre clinical setting and demonstrating their practical utility in intrapartum risk stratification.^[^27]

Study 13: Mode of Delivery After One Caesarean Section (1991)

Reference: Steer PJ et al. *Mode of delivery after one caesarean section: audit of current practice in a health region*. British Journal of Obstetrics and Gynaecology 1991 (BJOG)^[^29]

Dataset: 1,059 women delivering at 15+ units in the North West Thames region (SMMIS), 1988.

Key Findings

- 395 women (37%) underwent elective repeat caesarean section; 664 (63%) were offered a trial of labour.[²⁹]
- 471 (71%) of those who laboured achieved a vaginal delivery — demonstrating that trial of labour is frequently successful.[²⁹]
- No significant correlation was found between the proportion of patients allowed to labour in a given unit and the success rate of trial of labour, indicating that **unit-level culture** rather than patient selection drives VBAC rates.[²⁹]
- Units allowing a **longer duration of labour** showed a trend towards greater success rates ($P < 0.05$).[²⁹]
- Both elective and emergency intrapartum caesarean sections were associated with significantly higher postnatal infection rates than vaginal delivery (14.7% and 16.0% vs 3.4%, respectively).[²⁹]
- The study challenged the prevailing view that repeat caesarean was always preferable, supporting a more liberal approach to allowing women a trial of labour.[²⁹]

Study 14: Fetal Growth Potential and Neonatal Outcome (1992)

Reference: Danielian PJ, Allman AC, Steer PJ. *Is obstetric and neonatal outcome worse in fetuses who fail to reach their own growth potential?* British Journal of Obstetrics and Gynaecology 1992; 99(6):452–454.[³⁰]

Design: Retrospective analysis, 197 singleton cephalic pregnancies at term, Chelsea and Westminster Hospital.

Key Findings

- Fetuses whose birthweight was **below the centile expected** from their third-trimester ultrasound scan (i.e., who failed to reach their own customised growth potential) had significantly worse obstetric and neonatal outcomes than those who tracked or exceeded predicted growth.[³⁰]
- **In-utero fetal growth trajectory** was as important for perinatal outcome as being small for gestational age per se — a distinction later formulated as "customised growth potential". [³⁰]
- This was an early empirical validation from Steer's group of the concept that population-based birth weight charts fail to capture individually important growth restriction in fetuses whose constitutional growth potential is above the population median.[³⁰]

Cross-Cutting Themes and Synthesis

Haemoglobin and Fetal Growth

Across multiple SMMIS studies, Steer and colleagues consistently demonstrated that the relationship between maternal Hb and fetal outcomes is **non-linear and U-shaped**. Both extremes of Hb — severe anaemia and failure of physiological haemodilution — are independently harmful to birth weight and perinatal survival. This work challenged the binary classification of "anaemia" and established that optimal Hb in pregnancy (95–105 g/L) is lower than traditional thresholds suggest.^{[7][13][^15]}

Ethnicity and Obstetric Risk

A persistent theme across the SMMIS research is the **differential risk profile across ethnic groups**, consistently demonstrating that Black and South Asian women face higher rates of preterm birth, meconium-stained amniotic fluid, gestational diabetes, and adverse neonatal outcomes — not all of which are explained by socioeconomic or BMI differences. These findings have been foundational in highlighting structural ethnic health inequalities in UK maternity care.^{[7][16][24][25][^5]}

Multi-Factor Intrapartum Risk

The later SMMIS publications (Studies 11–12) converge on a major clinical message: **single-parameter intrapartum monitoring is inadequate**. Fetal heart rate interpretation must be contextualised by meconium, maternal pyrexia, FGR, tachysystole, nulliparity, oxytocin use, and ethnicity. The risk of adverse outcome with combined meconium and abnormal FHR is 15 times higher than with neither factor alone.^{[27][5]}

Blood Pressure and Perinatal Mortality

The SMMIS-based blood pressure study (Study 3) was ground-breaking in demonstrating that **low maternal blood pressure in pregnancy is as hazardous as hypertension** — a counterintuitive finding that had been largely overlooked in obstetric practice. Over 11% of perinatal deaths were attributable to blood pressures deviating from the physiological optimum (~83 mmHg diastolic), with most excess deaths occurring on the low-pressure side.^[^3]

Preterm Birth and the Environment

Two studies used the SMMIS data specifically to interrogate **environmental triggers of preterm birth** in London. The seasonal analysis confirmed a winter peak in preterm births unique to London, while the environmental pollution study ruled out short-term air quality and weather changes as proximate causes — thereby pointing to longer-term biological or socioeconomic mechanisms.^{[19][22]}

Summary Table of SMMIS Studies by Phil Steer

Publication Year	Study Topic	Sample Size	Journal	Key Result
1991	Mode of delivery after previous CS [²⁹]	1,059	BJOG	71% vaginal delivery with trial of labour
1992	Fetal growth potential and neonatal outcome [³⁰]	197	BJOG	Growth trajectory matters as much as size
1995	Maternal Hb and birth weight by ethnicity [⁷]	153,602	BMJ	Optimal lowest Hb = 95–105 g/L across all groups
2000	Maternal Hb and birth weight (review) [¹⁵]	222,614	Am J Clin Nutr	Severe anaemia and failure of haemodilution both harmful
2004	Gestation and ethnicity [¹⁶]	122,415	Int J Epidemiol	Blacks and Asians: shorter gestation, earlier maturity
2004	Maternal blood pressure and perinatal mortality [³]	210,814	BMJ	Low BP as dangerous as high BP; optimal ~83 mmHg
2005	Maternal Hb and perinatal mortality [¹³]	222,614	AJOG	U-shaped relation; lowest mortality at Hb 90–110 g/L
2006	Seasonal preterm birth [¹⁹]	482,765	BJOG	Single winter peak in London; 10% higher risk
2008	Air pollution/weather and preterm birth [²²]	482,568	Environ Res	No association with ambient pollution or weather
2011	Meconium: characteristics and risk factors [²⁵]	499,096	Obstet Gynecol	Black women 8× higher MSAF risk; gestational age key
2012	GDM, age, BMI, and ethnicity [²⁴]	174,320	BJOG	Black/Asian women at risk of GDM at lower age/BMI
2023	Multi-factor intrapartum risk and neonatal outcome [⁵]	302,137	BJOG	Meconium + aFHRp = 15× composite adverse outcome risk
2023	Interaction of intrapartum risk factors (case-control) [²⁷]	267	BJOG	Combined risk score outperforms single-factor assessment

Clinical and Policy Impact

Phil Steer's SMMIS-based research has influenced several areas of clinical practice and policy:

- **Haematology in pregnancy:** The U-shaped relationship between Hb and birth weight challenged WHO anaemia thresholds and contributed to guidance that optimal Hb in pregnancy is lower than commonly assumed.^{[7][15]}
- **Blood pressure monitoring:** Demonstrating harm from low maternal BP reinforced the importance of antenatal BP measurement across its full range — not just screening for hypertension.^[3]

- **Intrapartum monitoring:** The findings on combined meconium and FHR abnormalities directly informed NICE intrapartum care guidance on the need for multi-factor risk assessment, contributing to UK guidance that FHR interpretation must incorporate clinical context.^{[27][5]}
- **Ethnic health inequalities:** Consistently identifying heightened risk in Black and South Asian women has contributed to growing recognition of ethnic disparity in UK maternity outcomes and debate about ethnicity-specific screening thresholds for GDM.^{[16][24][^25]}
- **Seasonal service planning:** The winter peak in preterm births has practical implications for neonatal intensive care unit capacity and staffing in London.^{[19][20]}

References

1. [\[PDF\] Hemoglobin concentration in pregnancy and perinatal mortality: A ...](#) - KEY WORDS. Objective: Failure of fetal growth during pregnancy, and preterm birth, are the major cau...
2. [\[PDF\] Risk factors for a serious adverse outcome in neonates](#) - 12. The 2015 Each Baby Counts report identified an average of six risk factors per labour associated...
3. [Maternal blood pressure in pregnancy, birth weight, and ...](#) - Objective To investigate the relation of diastolic blood pressure in pregnancy with birth weight and...
4. [Risk factors for a serious adverse outcome in neonates - PubMed](#) - A range of risk factors, including suspicion of fetal growth restriction, maternal pyrexia and prese...
5. [Risk factors for a serious adverse outcome in neonates - PMC - NIH](#) - A range of risk factors, including suspicion of fetal growth restriction, maternal pyrexia and prese...
6. [Comparing the odds of postpartum haemorrhage in planned home ...](#) - The strengths and limitations of the data. The SMMIS database is extremely useful for the study of p...
7. [Relation between maternal haemoglobin concentration and birth ...](#) - Some ethnic groups have higher rates of low birth weight and preterm delivery than white women, and ...
8. [Relation between maternal haemoglobin concentration and birth ...](#) - Some ethnic groups have higher rates of low birth weight and preterm delivery than white women, and ...
9. [\[PDF\] WITN6977005_0001 - Infected Blood Inquiry](#) - We used it to examine the relation between the lowest haemoglobin concentration in pregnancy, birth ...
10. [Factors influencing relative weights of placenta and newborn infant](#) - # Maternal haemoglobin and blood pressure should have been regarded as continuous variables {#articl...
11. [Philip Steer - Google Scholar](#) - Philip Steer. Porfessor of obstetrics Imperial College London. Verified email at imperial.ac.uk. mat...
12. [Birth weight: nature or nurture? - PubMed](#) - Objective: To investigate the relative role of environmental and genetic factors in the determinatio...
13. [Hemoglobin concentration in pregnancy and perinatal mortality](#) - The objective of the study was to determine the association between maternal hemoglobin concentratio...

14. [\(PDF\) Hemoglobin concentration in pregnancy and perinatal mortality](#) - This study investigates the association between maternal hemoglobin (Hb) levels during pregnancy and...
15. [Maternal hemoglobin concentration and birth weight](#) - PubMed - Severe anemia (hemoglobin <80 g/L) is associated with the birth of small babies (from both preterm l...
16. [Does gestation vary by ethnic group? A London-based ...](#) - PubMed - This research suggests that normal gestational length is shorter in Black and Asian women compared w...
17. [Does gestation vary by ethnic group? A London-based study of over 122 000 pregnancies with spontaneous onset of labour](#)
18. [Does gestation vary by ethnic group? A London-based study of over ...](#) - Abstract. Background Evidence exists that normal gestational length varies with ethnicity. This UK-b...
19. [Seasonal patterns and preterm birth: a systematic review ...](#) - PubMed - In London (SMMIS data set), however, preterm births peaked only once a year, in winter. Babies born ...
20. [Seasonal patterns and preterm birth: a systematic review of the ...](#) - Rates of preterm birth in developed countries were highest twice a year (once in winter and again in...
21. [Seasonal patterns and preterm birth: a systematic review of the ...](#) - A systematic review of the countries in which a seasonal pattern of pre term birth has been reported...
22. [A time-series analysis of any short-term effects of meteorological ...](#) - This large study, covering 13 years, suggests that there is no association between preterm births an...
23. [Publication Date](#) - A time-series analysis of any short-term effects of meteorological and air pollution factors on pret...
24. [An analysis of the interrelationship between maternal age, body ...](#) - Maternal age and BMI interact with racial group in relation to the prevalence of GDM. Both factors a...
25. [Maternal and fetal characteristics associated with meconium-stained ...](#) - Meconium-stained AF rates are different among races and across gestational age, and overall risk of ...
26. [Maternal and Fetal Characteristics Associated With Meconium ...](#) - Background: The aim of this study was to determine the risk factors for meconium-stained amniotic fl...
27. [A case-control study of the interaction of fetal heart rate ...](#) - PubMed - A case-control study of the interaction of fetal heart rate abnormalities, fetal growth restriction,...
28. [A case-control study of the interaction of fetal heart rate ...](#) - A case-control study of the interaction of fetal heart rate abnormalities, fetal growth restriction,...
29. [Mode of delivery after one caesarean section](#) - PubMed - Results: 395 (37%) women were delivered by elective repeat caesarean section and 664 (63%) were allo...
30. [Is obstetric and neonatal outcome worse in fetuses who fail to reach ...](#) - This study supports the hypothesis that in-utero fetal growth pattern is as important for perinatal ...